

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology
Subject: DSIP(01CT1513)	Aim: Gray-level transformation techniques
Open-ended-Problem: 07	Date:14-08-2026
	Enrollment No:92400133121

AIM:

- Five different degraded images are provided.
- Use suitable gray-level transformation techniques to enhance each image.
- After processing, describe your observations in detail for every enhancement step
- Explaining how the chosen operations improved the visual quality.
- Finally, write a clear conclusion summarizing the effectiveness and limitations of the applied gray- level techniques.

FIRST IMAGE:



PROBLEM:

- It is a low contrast image.
- The image details are not visible clearly.
- The image appears dull

Technique To Be Used:

- Contrast Stretching
 - It is a technique in which we try to expand the range of intensity values in an image.
 - Through this technique pixel values can be spread which will improve the image contrast.

Code:

pip install opencv-python

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```
#import libraries
```

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
#read the input image1
```

```
image1 = cv2.imread("images/Picture1.png")
if image1 is None:
    print("Error:Image not found")
    exit()
```

```
#display the input image1
```

```
plt.figure(figsize=(4,4))
plt.imshow(image1,cmap='gray')
plt.title("Input image")
plt.axis("off")
plt.show()
```

Output:

BEFORE TRANSFORMATION

Input image



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Code:

#CONSTRAST STRECHING

```
image1_float = image1.astype(np.float32)
```

```
low = np.percentile(image1_float,2)
```

```
high = np.percentile(image1_float,98)
```

```
stretched = ((image1_float-low) * 255 / (high - low))
```

```
stretched = np.clip(stretched,0,255).astype(np.uint8)
```

#display the output image

```
plt.figure(figsize=(4,4))
```

```
plt.imshow(stretched,cmap='gray',vmin=0,vmax=255)
```

```
plt.title("Stretched image")
```

```
plt.axis("off")
```

```
plt.show()
```

Output:

AFTER TRANSFORMATION

Stretched image



Observation:

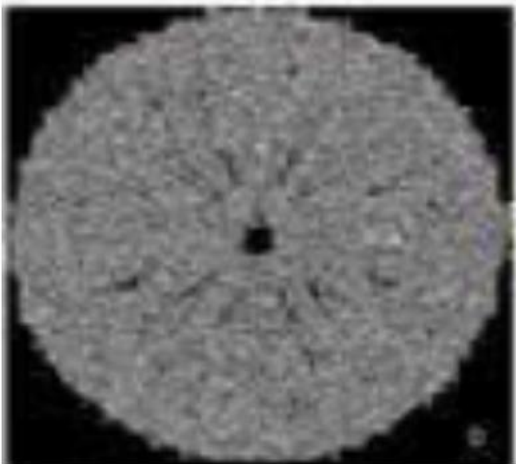
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- ✓ The intensity values are expanded over the complete range of grey levels.
- ✓ The image becomes brighter and more clear
- ✓ All the boundaries and edges are clearly defined
- ✓ The difference in foreground and background is improved
- ✓ Overall image quality is enhanced.

Conclusion:

- Contrast stretching effectively improved the image by using the full grey level range.
- This technique enhanced the visibility of edges and fine object details.
- Hence the image quality is enhanced.
- But if the image contains noise, then this technique may also enhance the noise along with the information.

SECOND IMAGE:



PROBLEM:

- Given image has poor visual contrast.
- Some internal structures are difficult to distinguish.

Technique To Be Used:

- Power Law Transformation

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- Gamma or power law transformation is a technique used to modify the brightness and contrast of image by applying non – linear transformation.
- The image will be transformed based on the gamma values
 - $\gamma > 1$ --dark
 - $\gamma < 1$ --brighter
 - $\gamma = 1$ —same

Code:

```
#read the input image2

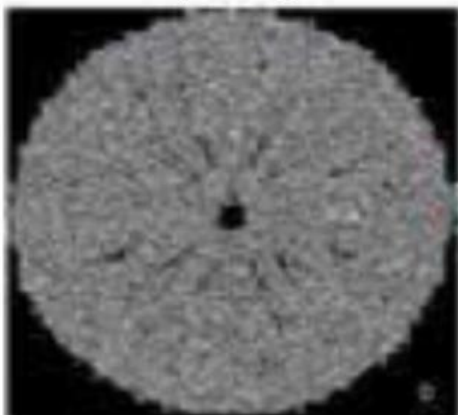
image2 = cv2.imread("images/Picture2.png")
if image2 is None:
    print("Error:Image not found")
    exit()
#display the input image2

plt.figure(figsize=(4,4))
plt.imshow(image2,cmap='gray')
plt.title("Input image")
plt.axis("off")
plt.show()
```

Output:

BEFORE TRANSFORMATION

Input image



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Code:

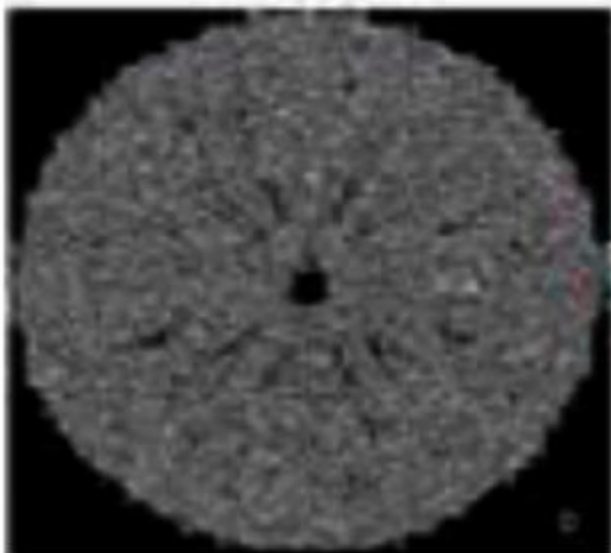
#power law

```
gamma = 1.5
gamma_image = np.array(255*(image2/255) ** gamma,dtype=np.uint8)
```

```
plt.figure(figsize=(4,4))
plt.imshow(gamma_image,cmap="gray")
plt.axis("off")
plt.show()
```

Output:

AFTER TRANSFORMATION



Observation:

- ✓ The gamma transformation changes the intensity distribution of image.
- ✓ The overall brightness is reduced compared to the original image.
- ✓ The enhancement is not accurate but moderate.
- ✓ Some internal structures are vaguely visible compared to the original image.

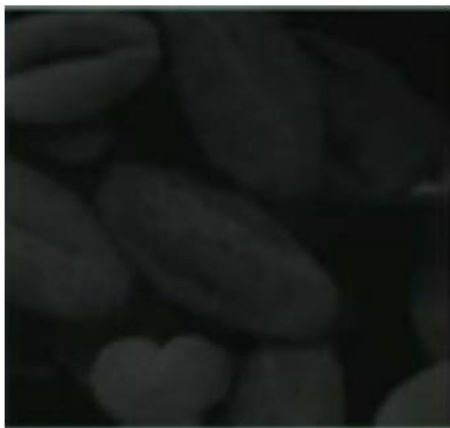
Conclusion:

- The gamma transformation provides moderate enhancement of given image.

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- It improves the image visibility of some internal variations
- However, the grainy structure could not be removed completely.
- This technique can not reduce the noise it can change the pixel intensities.

THIRD IMAGE:



PROBLEM:

- The given image is very dark image.
- It is little bit difficult to distinguish the object from background.
- Object details are not clearly visible.

Technique To Be Used:

- Power Law Transformation
 - Gamma or power law transformation is a technique used to modify the brightness and contrast of image by applying non – linear transformation.
 - The image will be transformed based on the gamma values
 - $\gamma > 1$ --dark
 - $\gamma < 1$ --brighter
 - $\gamma = 1$ —same

Code:

```
#read the input image3
```

```
image3 = cv2.imread("images/Picture3.png")
if image3 is None:
```

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```

print("Error:Image not found")
exit()
#display the input image3

plt.figure(figsize=(4,4))
plt.imshow(image3,cmap='gray')
plt.title("Input image")
plt.axis("off")
plt.show()

```

Output:
BEFORE TRANSFORMATION

Input image



Code:

```

#power law

gamma = 0.5
gamma_image = np.array(255*(image3/255) ** gamma,dtype=np.uint8)

plt.figure(figsize=(4,4))
plt.imshow(gamma_image,cmap="gray")
plt.axis("off")

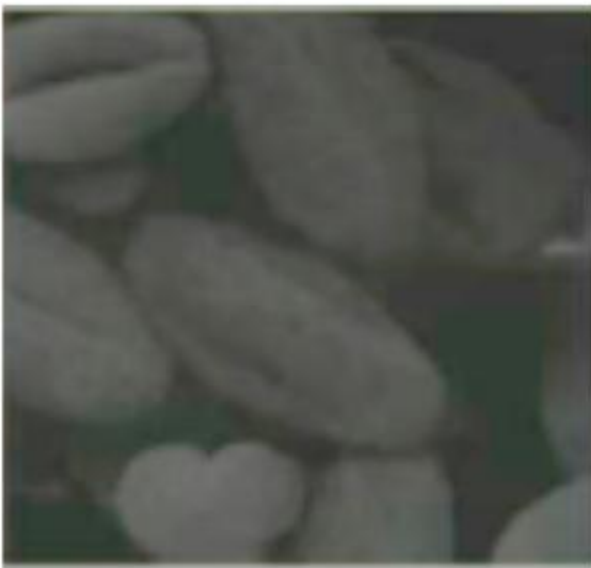
```

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plt.show()

Output:

AFTER TRANSFORMATION



Observation:

- ✓ The overall brightness of the image increased.
- ✓ The objects are clearly visible compared to the original image.
- ✓ Details in the darker regions become more visible.
- ✓ The transformation improves the visual quality without converting into binary image.

Conclusion:

- The gamma transformation enhanced the very dark image.
- It improves the image visibility of some internal variations
- The intensity levels are expanded.
- This technique can make brighter regions more saturated and reduce contrast in those regions.

FOURTH IMAGE:

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PROBLEM:

- The input image is severely underexposed
- The surrounding details are barely visible

Technique To Be Used:

- Log Transformation
 - This technique expands the lower intensity values making details present in dark regions more visible.
 - The log transformation is given by:

$$s = c\log(1+r)$$

Code:

```
#read the input image4

image4 = cv2.imread("images/Picture4.png")
if image4 is None:
    print("Error:Image not found")
    exit()
#display the input image4

plt.figure(figsize=(4,4))
plt.imshow(image4,cmap='gray')
plt.title("Input image")
plt.axis("off")
plt.show()
```

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Output:

BEFORE TRANSFORMATION

Input image



Code:

```
#apply log transformation
c = 80
log_image = c * np.log(1+image4.astype(np.float32))
log_image = np.array(log_image,dtype=np.uint8)

plt.figure(figsize=(4,4))
plt.imshow(log_image,cmap="gray")
plt.title("Log image")
plt.axis("off")
plt.show()
```

Output:

AFTER TRANSFORMATION

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Log image



Observation:

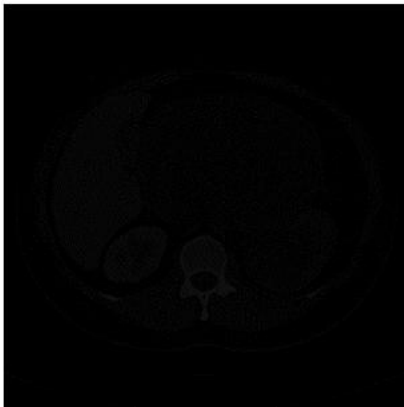
- ✓ The overall visibility of the dark image is improved.
- ✓ The car which was barely visible in the original image are also visible after the transformation.
- ✓ Even the outlines and details of car are also visible.
- ✓ Other than this the road, surroundings and street light are also revealed properly.
- ✓ But if there are any noisy variations, they will also be visible after enhancement which is a limitation of this transformation.

Conclusion:

- Log Transformation, this technique effectively enhanced the severely dark image by expanding the intensity levels.
- Most importantly it revealed the hidden details from the original image like car, street and surrounding scenes.
- However, this method may leave some noise variations noticeable.

FIFTH IMAGE:

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PROBLEM:

- The image is very dark which almost looks blank.
- Due to its poor intensity distribution the boundaries, fine details nothing are visible
- One cannot identify the internal structures in the given image.

Technique To Be Used:

- Log Transformation
 - This technique expands the lower intensity values making details present in dark regions more visible.
 - The log transformation is given by:

$$s = \text{clog}(1+r)$$

Code:

```
#read the input image5
```

```
image5 = cv2.imread("images/Picture5.png")
```

```
if image5 is None:
```

```
    print("Error:Image not found")
```

```
    exit()
```

```
#display the input image5
```

```
plt.figure(figsize=(4,4))
```

```
plt.imshow(image5,cmap='gray')
```

```
plt.title("Input image")
```

```
plt.axis("off")
```

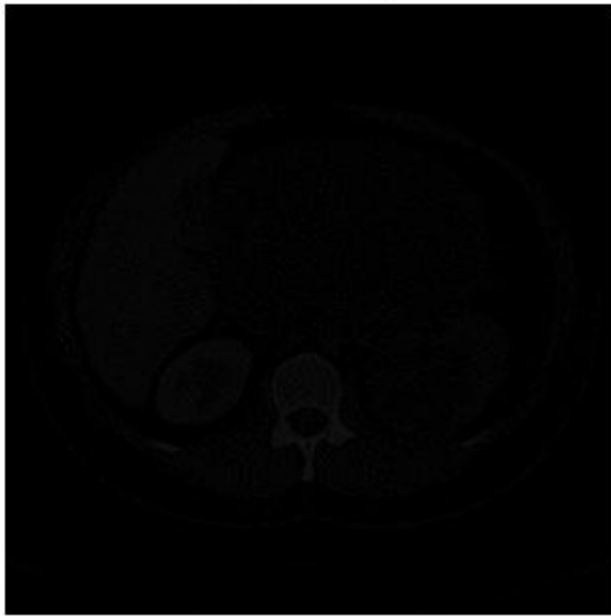
```
plt.show()
```

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Output:

BEFORE TRANSFORMATION

Input image



Code:

```
#apply log transformation
c = 80
log_image = c * np.log(1+image5.astype(np.float32))
log_image = np.array(log_image,dtype=np.uint8)

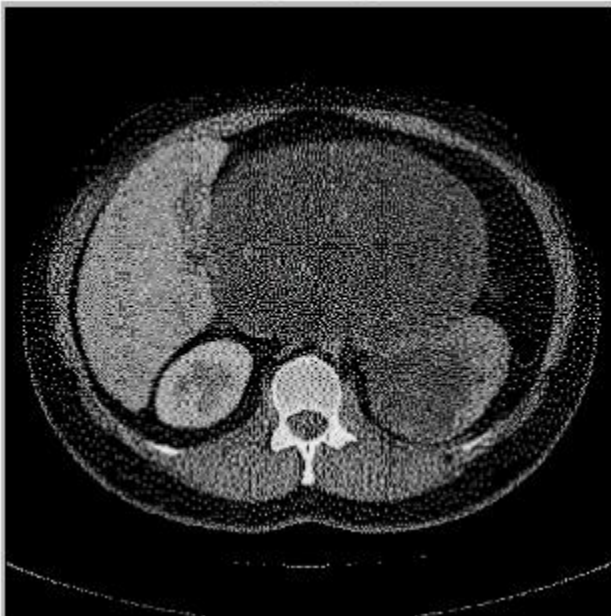
plt.figure(figsize=(4,4))
plt.imshow(log_image,cmap="gray")
plt.title("Log image")
plt.axis("off")
plt.show()
```

Output:

AFTER TRANSFORMATION

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Log image



Observation:

- ✓ The overall visibility of the medical scan is improved.
- ✓ The anatomical structures which are barely visible in the original image are clearly visible after the transformation.
- ✓ The boundaries of different regions can be properly identified
- ✓ Details present in the darker intensity are properly revealed.
- ✓ Some noise intensity variations are still noticeable.

Conclusion:

- Log transformation, this technique proved to be useful as it effectively enhanced the image.
- The lower intensity levels are expanded which significantly revealed the hidden details from the original image.
- However, there is a limitation in this technique, the noise variations can not be completely ignored they are also enhanced along with the information.